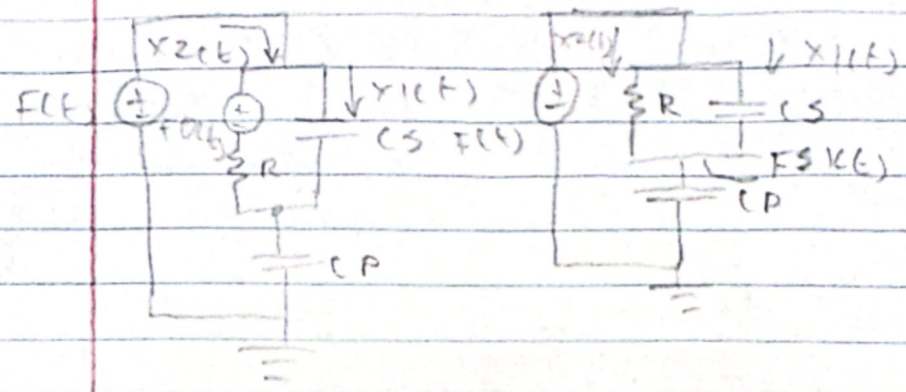
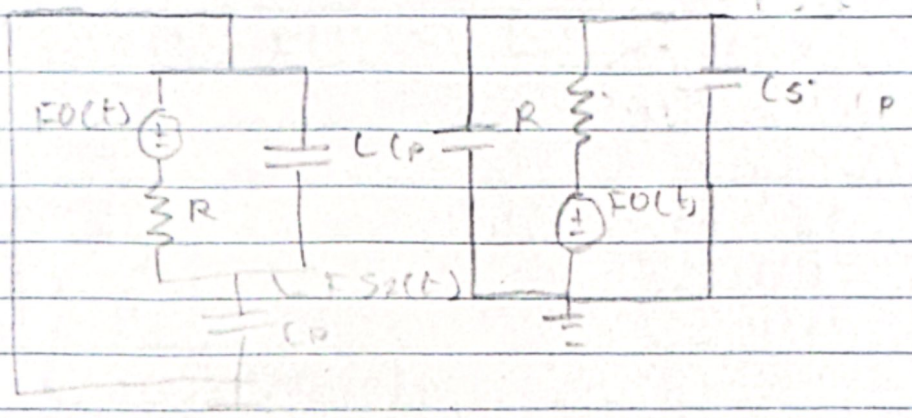
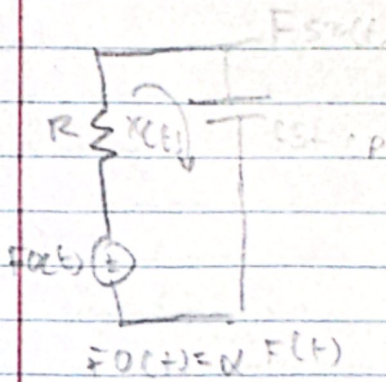






$$FS(t) = F_{S1}(t) + F_{S2}(t)$$

Análisis del circuito por superposición

Circuito 1 Apagando $E_0(t)$

$$X(t) = X_1(t) + X_2(t) \qquad X(t) = C_p \frac{dF_{S1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{S1}(t)]}{dt}$$

$$X_2(t) = \frac{E(t) - F_{S1}(t)}{R}$$

$$C_p \frac{dF_{S1}(t)}{dt} = C_s \frac{d[F(t) - F_{S1}(t)]}{dt} + \frac{E(t) - F_{S1}(t)}{R}$$

$$C_p S F_{S1}(S) = C_s [F(S) - F_{S1}(S)] + \frac{F(S) - F_{S1}(S)}{R}$$

$$C_p S F_{S1}(S) = C_s S F(S) - C_s S F_{S1}(S) + \frac{F(S)}{R} - \frac{F_{S1}(S)}{R}$$

$$\left(C_p S + C_s S + \frac{1}{R}\right) F_{S1}(S) = \left(C_s S + \frac{1}{R}\right) F(S)$$

$$\left(\frac{R(C_p + C_s)S + 1}{R}\right) F_{S1}(S) = \frac{C_s R S + 1}{R} F(S)$$

$$F_{S1}(S) = \frac{\frac{C_s R S + 1}{R} F(S)}{\frac{R(C_p + C_s)S + 1}{R}} = \frac{C_s R S + 1}{R(C_p + C_s)S + 1} F(S)$$

Circuito 2: Apagando $F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\,dt$$

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$F(s) = \frac{X(s)\left[R(C_p + C_s)s + 1\right]}{\alpha (C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{\frac{X(s)\left[R(C_p + C_s)s + 1\right]}{-\alpha}} = \frac{-\alpha}{R(C_p + C_s)s + 1}$$

$$F_2(s) = \frac{-\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F(s) = F_{s1}(s) + F_2(s) = \frac{(C_s R s + 1) F(s)}{R(C_p + C_s)s} - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} \qquad \frac{E_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

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